Solutions and teaching notes for chapter 4

Teacher use. The answers below concern the tasks not worked in the lesson. First have students state their method; give graded help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. $\dfrac{8-2}{3-1} = 3$. *Action:* Represent the horizontal and vertical increments.
2. $4h + h^2$. *Action:* Expand before simplifying.
3. $4 + h$. *Action:* Factor h out of the numerator before dividing.
4. 0 and 0.1. *Action:* Check by squaring.
5. 3 and 2. *Action:* Represent the distance to 0.
6. $y = 2x + 3$. *Action:* Substitute the coordinates to find the y-intercept.

Investigation activity

The slopes are $2 + h$; limit 2. The line passes through $(1, 1)$, so $y = 2x - 1$. The drawing motivates, the calculation justifies.

Immediate applications

1. $f(2 + h) - f(2) = 7h + h^2$; rate $7 + h$ for $h \neq 0$.
2. The rate is $4 + 3h$, so the derivative at 1 is 4.
3. $y = 5 - 2(x - 3) = -2x + 11$; approximation $5 - 2(0.04) = 4.92$.
4. The rate at 2 is $\dfrac{|h|}{h}$, with two opposite limiting values. For the square root at 0, the rate $\dfrac{1}{\sqrt{h}}$ for $h > 0$ exceeds every bound as h gets closer to 0: no real derivative.

Exercises: method and results

Exercise 4.1 – Two secants. Slopes 3 and 2.1; derivative at 1 is 2. The second is closer to the slope of the tangent; this is an exact calculation of a rate.

Exercise 4.2 – A tangent. Rate $4a + 2h = -4 + 2h$, so $f\prime(-1) = -4$. $f(-1) = 3$; tangent $y = 3 - 4(x + 1) = -4x - 1$.

Exercise 4.3 – Marginal cost. $C(10) = 400$, $C(11) = 441$, increase 41 for one unit. The rate around 10 is $40 + h$, so derivative 40: an approximation of the increase for a small change, not an equality for a whole unit.

Exercise 4.4 – Reading a tangent. On the figure in the lesson: points (1;0) and (2;4), slope 4. In the new situation, $f(2) = 3$, slope $-\dfrac{1}{2}$. Approximation $3 - 0.5 \times 0.1 = 2.95$. The y-coordinate of B is not necessarily $f(4)$.

Exercise 4.5 – Checking an approximation. $1 + 2(-0.02) = 0.96$. Exact 0.9604; exact minus approximate $0.0004 > 0$. The omitted term is h^2: approximation from below.

Exercise 4.6 – Tangent and intersection. The tangent has equation $y = 1 + 3(x-1) = 3x - 2$. Expanding the product indeed gives $x^3 - 3x + 2$. The x-coordinates of the intersections are therefore 1 and -2, by the zero-product rule. The points are $(1,1)$ and $(-2,-8)$. There are two common points: that is enough to refute the general claim.

Exercise 4.7 – Numerical secants. List of rates $2 + h$: 2.1; 2.01; 2.001 and 1.9; 1.99; 1.999. The algebraic calculation gives the real number 2. For very small h, floating-point calculation can lose precision.

Exercise 4.8 – Delayed review. For $h \neq 0, -1$, the rate is $-\dfrac{1}{1+h}$; it tends to -1. Approximation $1 - 0.02 = 0.98$. The exact value $\dfrac{50}{51}$ allows a check.

CQFD, exercise 17 p. 127, all questions. The x-coordinates of A, B and C are 0, 1 and 2. The tangents give slopes 2, 0 and -2 respectively. Thus $f\prime(0) = 2$, $f\prime(1) = 0$, $f\prime(2) = -2$. The units of the axes must be read from the grid.

CQFD, exercise 19 p. 127, questions 1 and 2. The point A reads $(1,4)$, so $f(1) = 4$. The tangent passes through $(0,1)$ and $(1,4)$: slope 3. Thus $f\prime(1) = 3$ and the tangent has equation $y = 4 + 3(x-1) = 3x + 1$.

MCQ: answers and reasons

MCQ 1.

1. B. $\dfrac{4-1}{2-1} = 3$.
2. A. $4 + 2(x-1)$.
3. D. It is a value at a point.
4. C. Rate $6 + h$.
5. B. Tangent affine function.

MCQ 2.

1. C. Rate $\dfrac{1}{\sqrt{h}}$ not finite.
2. A. $|h| = -h$.
3. D. Constant y-coordinate.
4. B. A change in position divided by a change in time is expressed in $\mathrm{m\,s^{-1}}$.
5. C. Exact equality for an affine function, approximation in general.

True or false

1. False: x^3 and the tangent $3x - 2$ at 1 also meet at -2.
2. False: the rates on either side are -1 and 1.
3. True: the rate is exactly m for every nonzero h.

Differentiated teaching. For a technical block, offer only the first step (domain, factor or reference sample space), then withdraw the help. For an independent student, ask for a second method and the treatment of a limiting case. Orally, allow an initial reformulation in the language the student has mastered, then require a precise mathematical sentence in French.